TEAM 1 RAG AGENT · HOW PAPERS ARE INDEXED

How a paper section becomes search chunks

Real examples from our index. Each PDF is split into sections by its headings, each section into chunks of at most 1,200 characters, and every chunk carries the paper and section name so it can be found and cited on its own.

39	2,983	6,455	835
papers indexed	sections (parents)	chunks (children)	median chunk length, characters

THE PIPELINE

Two levels: small chunks to search, whole sections to answer from

AT INDEXING TIME

PDF layout-aware text → **Sections** split at headings; figure captions kept apart → **Chunks** ≤ 1,200 chars incl. name prefix →

Chroma embeddinggemma vectors

AT QUESTION TIME

Search vectors + keywords, 24 candidates → **Rerank** MiniLM; drop scores below 1.5 →

Expand chunk → its section, ≤ 2,400 chars → **Gemma** up to 5 sections, ≤ 14,000 chars

Chunks are small so a search hit is precise. Gemma does not see the chunk alone: retrieval swaps each hit for a window of its whole section, so the answer has the surrounding sentences, and only one window per section is sent.

EXAMPLE 1 · SPLIT AT PARAGRAPH BREAKS

“1 Introduction” of *Attention Is All You Need*

The section is 1,924 characters, too long for one chunk, so the splitter cuts it into three. It tries separators in order — blank line, line break, sentence end, space — and here every cut

falls on a blank line between paragraphs. Chunk 2 holds two paragraphs because together they still fit.

chunk 1 chunk 2 chunk 3

1 Introduction

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [35, 2, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [38, 24, 15].

Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states ht, as a function of the previous hidden state ht−1 and the input for position t. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [21] and conditional computation [32], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduc- tion models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 19]. In all but a few cases [27], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

Chunk 1 496 / 1,200

```
Paper: 03_Attention_Is_All_You_Need.pdf
Section: 1 Introduction
```

1 Introduction

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks … 212 more characters … continued to push the boundaries of recurrent language models and encoder-decoder architectures [38, 24, 15].

Chunk 2 1,169 / 1,200

```
Paper: 03_Attention_Is_All_You_Need.pdf
Section: 1 Introduction
```

Recurrent models typically factor computation along the symbol positions of the input and output sequences. Al … 885 more characters … all but a few cases [27], however, such attention mechanisms are used in conjunction with a recurrent network.

Chunk 3 447 / 1,200

```
Paper: 03_Attention_Is_All_You_Need.pdf
Section: 1 Introduction
```

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirel … 163 more characters … state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

> Each chunk starts with `Paper: …` and `Section: …`. A chunk that only says “This inherently sequential nature precludes parallelization” would not match a question about the Transformer paper by itself; the prefix ties it to the paper and section for search and for the citation.

EXAMPLE 2 · OVERLAP INSIDE A PARAGRAPH

“2 Related Work” of the DPO paper, chunks 3 and 4

When a paragraph is longer than the space left, the cut falls mid-paragraph. Then the next chunk repeats the end of the previous one, up to 120 characters, so a sentence split across the boundary is still whole in at least one chunk. Here the two chunks share **104** characters.

Chunk 3 · end 1,170 / 1,200

… rs [49]. Similarly, preference-based RL (PbRL) learns from binary preferences generated by an unknown 'scoring' function rather than rewards [9, 37]. Various algorithms for PbRL exist, including methods that can reuse off-policy preference data, but generally involve first explicitly estimating the latent scoring function (i.e. the reward model) and subsequently

Chunk 4 · start 328 / 1,200

```
Paper:
10_DPO_Direct_Preference_Optimization.p
df Section: 2 Related Work
```

involve first explicitly estimating the latent scoring function (i.e. the reward model) and subsequently optimizing it [16, 9, 12, 36, 21]. We instead present a single stage policy learning approach that directly optimizes a policy to satisfy preferences.

Overlap is the exception, not the rule: **19%** of neighbouring chunk pairs (655 of 3,472) share text, with a median of 93 characters. Most cuts fall on paragraph breaks, where no overlap is needed.

EXAMPLE 3 · FIGURE CAPTIONS

Captions are their own short chunks

Figure and table captions become separate chunks marked as captions (881 of the 6,455). The agent may quote a caption's words but never claims to have read the figure or table itself.

page 3: Figure 1 105 / 1,200

```
Paper: 03_Attention_Is_All_You_Need.pdf
Section: Figure 1
```

Figure 1: The Transformer - model architecture.

ACROSS THE WHOLE INDEX

Most sections fit in one chunk

Of 2,983 sections, 1,921 (64%) are a single chunk. Long sections such as related work, experiments and reference lists make up the tail.



Chunk length (characters, prefix included)	Value
Shortest (a one-line caption)	42
Median	835
90th percentile	1,156
Longest (the limit)	1,200

SETTINGS

Where each number comes from

Setting	Value	Where
Chunk size limit, prefix included	1,200 chars	`MAX_CHUNK_CHARACTERS` · `app/config.py`
Overlap when a cut falls inside a paragraph	120 chars	`CHUNK_OVERLAP_CHARACTERS` · `app/config.py`
Separators, tried in order	`\n\n` · `\n` · `". "` · `" "`	`split_sections_into_chunks` · `app/document_processing.py`
Candidates per search	24	`CANDIDATE_COUNT` · `app/retrieval.py`
Reranker minimum score	1.5	`MIN_RERANK_SCORE` · `.env`
Section window per hit	2,400 chars	`expand_parent_section_windows` · `app/retrieval.py`
Sections sent to Gemma / total budget	5 / 14,000 chars	`EVIDENCE_COUNT`, `CONTEXT_CHARACTERS` · `app/retrieval.py`

Changing the chunk size or overlap changes the index signature: the app then refuses to start until ingestion is run with `--rebuild`.

Numbers and text read from the Chroma index on 1 Nov 2026 (`data/chroma`, collections `research_papers` and `research_sections`).